PHY 2506
Problem Set 2
Assigned Oct 16th; Due Nov 13th, 2026

1. Obtain the two sample retrievals of the vertical profile of CO from the Tropospheric Emission Spectrometer (TES), "TES_prof_sample1.dat" and "TES_prof_sample2.dat," and the corresponding files with the averaging kernels and covariance matrices. (If you are using Python, you can use the included file "pset2_ques1.ipynb" to read these data files.) The contents of the data files are described in the included file "Notes_for_problem1_pset2.pdf." The TES retrieval can be described by the equation

$$\hat{\mathbf{x}} = \mathbf{x}^b + \mathbf{A}(\mathbf{x} - \mathbf{x}^b) = (\mathbf{I} - \mathbf{A})\mathbf{x}^b + \mathbf{A}\mathbf{x} \quad (1)$$

where $\mathbf{A}$ is the TES averaging kernel and $\mathbf{x}^b$ is the a priori profile used in the retrieval. For TES, the retrieval is performed with respect to the natural logarithm of the volume mixing ratio (VMR) of the trace gas. Therefore, the quantities $\hat{\mathbf{x}}$, $\mathbf{x}$, and $\mathbf{x}^b$ should be expressed in terms of the natural log of the VMR. In interpreting the averaging kernel you may find it helpful to know that the TES retrievals are performed in the presence of clouds, but for clouds of large optical depths the retrievals can only extend down to the cloud top.

(a) Plot the first 20 rows of the averaging kernel matrices for the two retrievals of CO. The y-axis should be log-pressure and plot only from 1000 hPa to 10 hPa (with log-pressure on the y-axis, in decreasing order). Calculate the number of independent pieces of information obtained in the retrievals.

(b) Calculate and plot the three leading eigenvectors and eigenvalues of the averaging kernel matrices. How are the eigenvalues related to the number of independent pieces of information in the retrievals that you calculated in Part (a)?

(c) Calculate and plot the three leading eigenvectors and eigenvalues of the observation error covariance matrices. What fraction of the total variance is captured by these three modes? How is the first mode different for the two retrievals?

(d) Obtain the a priori profile ("mopitt_xa.dat") used in the MOPITT retrievals of CO. Since the TES retrieval is given by Equation (1), we can substitute the a priori profile used in the two TES retrievals with the MOPITT a priori profile to transform the TES retrievals as follows:

$$\hat{\mathbf{x}}_{trans} = \hat{\mathbf{x}} - (\mathbf{I} - \mathbf{A})\mathbf{x}^b + (\mathbf{I} - \mathbf{A})\mathbf{x}^b_{MOP} \quad (2)$$

where $\mathbf{x}^b_{MOP}$ and $\hat{\mathbf{x}}_{trans}$ are the MOPITT a priori profile and the transformed TES CO retrieval, respectively. The trace gas abundances in the retrieval (Equations (1) and (2)) must be expressed as the natural log of the VMR and recall that 1 ppb is a VMR of $1\text{x}10^{-9}$. The transformed retrieval reflects the influence of the MOPITT a priori profile. For each of the TES retrievals, plot the original retrieved and a priori CO profiles (only plot the levels between the surface and 10 hPa), the MOPITT a priori, and the transformed TES

CO profile that you obtained using the MOPITT a priori. Comment on the differences between the original and transformed retrieved profiles, and the differences between the retrieved and a priori profiles.

2. Starting from the expression for the averaging kernel $\mathbf{A} = \mathbf{KH}$, where $\mathbf{K}$ is the gain matrix and $\mathbf{H}$ is the Jacobian (or observation operator/forward model), relate the eigenvectors and eigenvalues of the averaging kernel $\mathbf{A}$ to the singular vectors and singular values of the normalized Jacobian $\tilde{\mathbf{H}}$.

3. In this problem you will use observations of atmospheric carbon monoxide (CO) from the Measurement Of Pollution In The Troposphere (MOPITT) satellite instrument and data from the GEOS-Chem chemical transport model to estimate the emissions of CO from regions within Asia. The optimal estimate of the sources of CO is obtained by minimizing the MAP cost function [*Rodgers*, 2000]

$$J(\mathbf{x}) = [\mathbf{y} - H(\mathbf{x})]^T \mathbf{R^{-1}} [\mathbf{y} - H(\mathbf{x})] + (\mathbf{x} - \mathbf{x}^b)^T \mathbf{B^{-1}} (\mathbf{x} - \mathbf{x}^b) \quad (3)$$

where $\mathbf{y}$ is the observation vector whose elements are the MOPITT observations of the column abundance of CO, $\mathbf{R}$ is the observation error covariance matrix, $\mathbf{x}$ is the state vector whose elements are the strengths of the sources of CO for the different regions in Asia shown in Figure 1, and $H(\mathbf{x})$ is the observation operator that transforms the CO sources into concentrations of CO (through the GEOS-Chem model and accounts for the vertical sensitivity of MOPITT and the a priori constraints used in the MOPITT retrievals). The quantity $\mathbf{x}^b$ is the a priori state vector (the a priori CO emissions) and $\mathbf{B}$ is the a priori error covariance matrix, which we assume to be diagonal (i.e. that the sources are uncorrelated). The expression for the observation operator is

$$H(\mathbf{x}) = \mathbf{y}^b_{MOP} + \mathbf{A}_{yy}(M(\mathbf{x}) - \mathbf{y}^b_{MOP}) \quad (4)$$

where $M(\mathbf{x})$ represents the GEOS-Chem model, $\mathbf{y}^b_{MOP}$ is the MOPITT a priori constraint and $\mathbf{A}_{yy}$ is the MOPITT averaging kernel matrix. We also assume that the individual MOPITT column observations are uncorrelated, so $\mathbf{R}$ is diagonal.

For Gaussian statistics, with a linear observation operator, the a posteriori estimate of the state vector is obtained by

$$\mathbf{x}^a = \mathbf{x}^b + (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H} + \mathbf{B}^{-1})^{-1} \mathbf{H}^T \mathbf{R}^{-1} (\mathbf{y} - \mathbf{H}\mathbf{x}^b) \quad (5)$$

where $\mathbf{H} = \partial H(\mathbf{x}) / \partial \mathbf{x}$ is the Jacobian matrix. Using Equation (4), the Jacobian can be expressed as

$$\mathbf{H} = \frac{\partial H(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{A}_{yy} \frac{\partial M(\mathbf{x})}{\partial \mathbf{x}} . \quad (6)$$

The quantity $\partial M(\mathbf{x}) / \partial \mathbf{x}$ is the sensitivity of the modeled concentration of CO to the sources, and is calculated by using the tagged tracer simulation mode in GEOS-Chem.

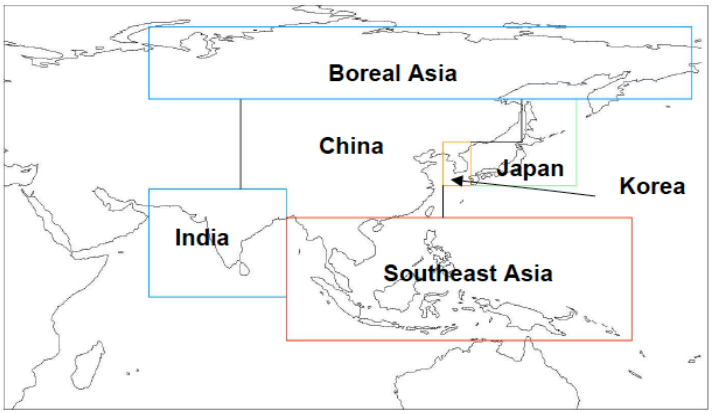


Figure 1. Regional definition of the seven-element state vector, which consists of the six regions shown here plus the CO emissions from the rest of the world (ROW).

For this problem you will be provided with MOPITT CO column data and GEOS-Chem output for March 1-27, 2001, and MATLAB code to estimate the a posteriori CO emissions using Equation (5). The objective is to understand the sensitivity of the source estimates to your assumptions about the observation error covariance, the a priori error covariance, and the overall inversion framework. A good reference for this problem is *Heald et al*. [2004], which provides a benchmark against which you can compare your analysis.

(a) Run the inversion code inverseCO.m. The required data files are in the "data" folder. You will be asked two questions (for the observation and a priori errors). Hitting "return" will give the default results. The code should print a table with the a posteriori and a priori source estimates, a table of the a posteriori correlation matrix, and the averaging kernel matrix for the inversion. It will also plot the rows of the averaging kernel matrix. For this default configuration, which source estimate is the least well constrained? What source elements are well constrained?

(b) What are the degrees of freedom for signal (DOFs) for this default setup? Add a line to the code to calculate this.

(c) How large are the differences between the a posteriori and a priori source estimates (the increments), relative to the a priori error? Are the increments larger than the priori error estimates for any of the source regions? If so, why might that have happened?

(d) Now run the code with the default observation error, but increase the a priori error to 0.9. How do the increments now compare to the a priori error?

i) How do the DOFS change, and why?

ii) What happened to the estimate that is the least well constrained?

(e) Now run the code with an observation error of 0.1 and an a priori error of 0.9. What are the DOFs? Do you think this is an accurate reflection of the sensitivity of the inversion to the CO emissions? (Hint: consider the value of the reduced χ^2)

(f) In your results in Part (e), which are the source regions for which the increments are larger than the a priori error? Considering the correlation matrix, can you explain why for these elements you obtained increments that are larger than the a priori error?

(g) Run the six-element inversion code (inverseCO_6x.m) using the default error values. What is the difference between the six-element and seven-element code? How do the source estimates compare to those obtained in Part (a)? Are the differences in the source estimates between these two inversions consistent with the a posteriori errors?

(h) Now run the six-element inversion code with an observation error of 0.1 and an a priori error of 0.9. Compare the source estimates obtained in this case to those from Parts (a) and (e). Comment on the quality of the source estimates.